Hands-On 02

LINEAR REGRESSION

1. Load the dataset (compressive strength) and inspect it (column names, data types, missing values, duplicate rows, and descriptive statistics). Report the number of duplicate rows and state, with a reason, whether you remove them before splitting. Define the features and the target, split the data 80:20 with random_state=42, and scale the features with StandardScaler fitted on the training set only. Do not use the test set again until the final evaluation

2. Apply Linear Regression, Ridge (L2), and Lasso (L1) with scikit-learn's default regularization strength (alpha=1.0).

3. Compare the MAE, RMSE, and R^2 of the three models on the test set. Also report training R^2 and comment on whether any model shows signs of overfitting or underfitting.

4. Display the learned coefficients of the Linear Regression model. Because the features are standardized, each coefficient is the change in predicted strength (MPa) for a one-standard-deviation increase in that feature, holding the others fixed.

5. For the best-performing model, plot predicted values against the true compressive strengths, and plot residuals against predicted values and against Age.

LOGISTIC REGRESSION

6. Load and inspect the alzheimers_disease_data dataset, including data types, missing values, duplicates, descriptive statistics, and the distribution of Diagnosis. Remove PatientID and DoctorInCharge. Split the data 80:20 using random_state=42 and stratify=y. Fit preprocessing only on the training data.

7. Apply unregularized Logistic Regression, L2 Logistic Regression, and L1 Logistic Regression using C=1.0 and max_iter=1000.

8. Compare test accuracy, precision, recall, F1-score, and ROC–AUC. Report training and test ROC–AUC and comment on possible overfitting or underfitting.

9. Display the coefficients and odds ratios of the unregularized model. Identify the five most influential features and explain whether each increases or decreases the predicted odds of Alzheimer's diagnosis.

10. For the best model, display the confusion matrix and plot the ROC and Precision–Recall curves. Briefly discuss its false positives and false negatives.

FINAL SUBMISSION - OPEN-ACCESS IPYNB LINK

- Run all cells and keep outputs visible; include all challenge code and answers.
- Select Share → General access → Anyone with the link → Viewer.
- Submit the Google Colab/IPYNB link in teams.
- Suggested filename: StudentID_HandsOn2.ipynb.

Important: A link that requests permission is not considered submitted.